MATH 533

REGRESSION AND ANALYSIS OF VARIANCE

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Assignment (i)

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1 Exercise 1

1.1 Part A

Set up: Assume we have the model $y = x\beta + e$ where x, y, e are random variables and β is fixed.

(i) TRUE: If $\mathbb{E}[e|x] = 0$, then the law of iterated expectation yields

$$\mathbb{E}[x^2 e] = \mathbb{E}[\mathbb{E}[x^2 e|x]] = \mathbb{E}[x^2 \mathbb{E}[e|x]] = \mathbb{E}[x^2 \cdot 0] = 0, \tag{1}$$

hence this will imply $\mathbb{E}[x^2|e] = 0$.

(ii) FALSE: If $\mathbb{E}[ex] = 0$, then we have

$$\mathbb{E}[x^2 e] = \mathbb{E}[x \cdot xe] = \mathbf{Cov}(x, xe) + \mathbb{E}[x]\mathbb{E}[xe] = \mathbf{Cov}(x, xe) \tag{2}$$

However we do not know the independence between x and xe and hence we cannot conclude $\mathbb{E}[x^2 e] = 0$.

(iii) FALSE: If $\mathbb{E}[e|x] = 0$, then

$$\mathbb{E}[xe] = \mathbb{E}[\mathbb{E}[xe|x]] = \mathbb{E}[x\mathbb{E}[e|x]] = 0 \tag{3}$$

and further

$$\mathbf{Cov}(x, e) = \mathbb{E}[xe] - \mathbb{E}[x]\mathbb{E}[e] = -\mathbb{E}[x]\mathbb{E}[e]. \tag{4}$$

If x, e are independent then $\mathbf{Cov}(x, e) = 0$. But from (4) there is no guarantee that x, e have mean zero, hence we can not conclude x, e are independent.

(iv) FALSE: If $\mathbb{E}[ex] = 0$, then

$$\mathbb{E}[ex] = \mathbb{E}[\mathbb{E}[ex|x]] = \mathbb{E}[x \cdot \mathbb{E}[e|x]] = 0 \tag{5}$$

however it does not imply $\mathbb{E}[e|x] = 0$.

(v) TRUE: If $[e, x]^\top$ is jointly Gaussian, $\mathbb{E}[e|x] = 0$, $\mathbb{E}[e^2|x] = \sigma^2$ (*I assume σ^2 here means the variance and it exists?*), then we know $\mathbb{E}[e] = 0$ and $\mathbb{E}[xe] = 0$, since

$$\mathbb{E}[e] = \mathbb{E}[\mathbb{E}[e|x]] = \mathbb{E}[0] = 0, \mathbb{E}[ex] = \mathbb{E}[\mathbb{E}[ex|x]] = \mathbb{E}[x\mathbb{E}[e|x]] = 0 \tag{6}$$

hence

$$\mathbf{Cov}(x, e) = \mathbb{E}[xe] - \mathbb{E}[x]\mathbb{E}[e] = 0 \tag{7}$$

We have the assumption that $[x, e]^\top$ is jointly Gaussian, plus their covariance is zero, hence x, e are independent.

1.2 Part B

If $\beta_1 = \gamma_1$, then the model can be described by

$$\begin{cases} y = x\gamma_1 + e \\ y = x\gamma_1 + x^2\beta_2 + u \end{cases} \tag{8}$$

which means we will have $e = x^2\beta_2 + u$. Note that we have the orthogonality property given by $\mathbb{E}[xe] = 0$, so we substitute e by $x^2\beta_2 + u$ and we have

$$\mathbb{E}[x(x^2\beta_2 + u)] = \mathbb{E}[\mathbb{E}[x(x^2\beta_2 + u)|x]] = \mathbb{E}[x^3\mathbb{E}[\beta_2|x] + x\mathbb{E}[u|x]] = 0 \tag{9}$$

since we already know that $\mathbb{E}[u|x] = 0$, so we indeed need to have $\mathbb{E}[x^3\mathbb{E}[\beta_2|x]] = 0$, i.e $\mathbb{E}[\beta_2|x] = 0$, that is the condition so that $\beta_1 = \gamma_1$.

2 Exercise 2

2.1 Part A

(1)

In this exercise, we have

$$\mathcal{O}_{LS}(\beta_1, \beta_2) = \sum_{i=1}^{n} \{y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta}\}^2 \tag{10}$$

$$= \arg\min_{\beta_0, \beta_1} \frac{1}{n} \sum_{i=1}^{n} \{y_i - (\beta_0 + \beta_1 x_i)\}^2 \tag{11}$$

$$= \arg\min_{\beta_0, \beta_1} \frac{1}{n} \sum_{i=1}^{n} \{y_i^2 + \beta_0^2 + 2\beta_0\beta_1 x_i + \beta_1^2 x_i^2 - 2y_i\beta_0 - 2\beta_1 y_i x_i\} \tag{12}$$

Further, y_i^2 is independent of β_0, β_1, and we use the fact that

$$\frac{1}{n}\sum_{i=1}^{n} x_i := \bar{x}_n \qquad \frac{1}{n}\sum_{i=1}^{n} y_i := \bar{y}_n \tag{13}$$

to obtain

$$\mathcal{O}_{LS}(\beta_1, \beta_2) := \arg\min_{\beta_0, \beta_1} \left\{ \boldsymbol{\beta}_0^2 + \frac{1}{n}\sum_{i=1}^{n} x_i^2 \beta_1^2 + 2\bar{x}_n \beta_0 \beta_1 - 2\bar{y}_n \beta_0 - \frac{2}{n}\sum_{i=1}^{n} x_i y_i \beta_1 \right\}. \tag{14}$$

Define a multivariate function $f(\beta_0, \beta_1)$ by

$$f(\beta_0, \beta_1) = \beta_0^2 + \frac{1}{n}\sum_{i=1}^{n} x_i^2 \beta_1^2 + 2\bar{x}_n \beta_0 \beta_1 - 2\bar{y}_n \beta_0 - \frac{2}{n}\sum_{i=1}^{n} x_i y_i \beta_1 \tag{15}$$

and the gradient of f is given by

$$\nabla f = \left(\frac{\partial f}{\partial \beta_0}, \frac{\partial f}{\partial \beta_1}\right) = \left(2\beta_0 + 2\bar{x}_n\beta_1 - 2\bar{y}_n \ , \ \frac{2}{n}\sum_{i=1}^{n} x_i^2 \beta_1 + 2\bar{x}_n \beta_0 - \frac{2}{n}\sum_{i=1}^{n} x_i y_i \right). \tag{16}$$

Set $\nabla f = 0$, we solve the system

$$\begin{cases} 2\beta_0 + 2\bar{x}_n\beta_1 - 2\bar{y}_n & = 0 \\ \frac{2}{n}\sum_{i=1}^{n} x_i^2\beta_1 + 2\bar{x}_n\beta_0 - \frac{2}{n}\sum_{i=1}^{n} x_i y_i & = 0 \end{cases} \tag{17}$$

where the first equation will directly yield

$$\widehat{\beta}_0 = \bar{y}_n - \bar{x}_n \widehat{\beta}_1. \tag{18}$$

Now use (18) to solve for $\widehat{\beta}_1$ in the second equation, we have

$$\frac{1}{n}\sum_{i=1}^{n} x_i^2 \widehat{\beta}_1 + \bar{x}_n(\bar{y}_n - \bar{x}_n\widehat{\beta}_1) - \frac{1}{n}\sum_{i=1}^{n} x_i y_i = 0 \tag{19}$$

which gives us

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}_n\bar{y}_n}{\sum_{i=1}^n (x_i - \bar{x}_n)^2}. \tag{20}$$

Note that (20) is just a reformation of the result in assignment, since one have

$$\begin{aligned}
\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n) &= \sum_{i=1}^n x_i y_i - \bar{y}_n \sum_{i=1}^n x_i - \bar{x}_n \sum_{i=1}^n y_i + n\bar{x}_n\bar{y}_n & (21)\\
&= \sum_{i=1}^n x_i y_i - n\bar{y}_n\bar{x}_n - n\bar{y}_n\bar{x}_n + n\bar{x}_n\bar{y}_n & (22)\\
&= \sum_{i=1}^n x_i y_i - n\bar{x}_n\bar{y}_n & (23)
\end{aligned}$$

and

$$\begin{aligned}
\sum_{i=1}^n (x_i - \bar{x}_n)^2 &= \sum_{i=1}^n x_i^2 + n\bar{x}_n^2 - 2\bar{x}_n \sum_{i=1}^n x_i & (24)\\
&= \sum_{i=1}^n x_i^2 + n\bar{x}_n^2 - 2\bar{x}_n^2 & (25)\\
&= \sum_{i=1}^n x_i^2 - n\bar{x}_n^2. & (26)
\end{aligned}$$

Since such $(\widehat{\beta}_0, \widehat{\beta}_1)$ is the unique point that cancels the gradient, followed by the fact that $\mathcal{O}_{LS}$ is strictly convex, so such $(\widehat{\beta}_0, \widehat{\beta}_1)$ is the unique minimizer, where

$$\widehat{\beta}_0 = \widehat{\beta}_0 = \bar{y}_n - \bar{x}_n\widehat{\beta}_1 \quad \text{and} \quad \widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{\sum_{i=1}^n (x_i - \bar{x}_n)^2}. \tag{27}$$

(2)

If only intercept is included, then it means $\boldsymbol{x}_i = (1, \cdots, 1)^\top$ hence

$$\mathcal{O}_{LS}(\boldsymbol{\beta}) = \arg\min_{\boldsymbol{\beta}\in\mathbb{R}^n} \frac{1}{n}\sum_{i=1}^n \left\{ y_i - \sum_{i=1}^n \beta_i \right\}^2. \tag{28}$$

Now for $\widehat{\beta}_n$, we simply have

$$\begin{aligned}
\arg\min_{\beta_0\in\mathbb{R}} \frac{1}{n}\sum_{i=1}^n \left\{ y_i - \sum_{i=1}^n \beta_i \right\}^2 &:= \arg\min_{\beta_0\in\mathbb{R}} \frac{1}{n}\sum_{i=1}^n \{y_i - \beta_n\}^2 & (29)\\
&= \arg\min_{\beta_0\in\mathbb{R}} \left\{ \frac{1}{n}\sum_{i=1}^n y_i^2 - \frac{2}{n}\sum_{i=1}^n y_i\beta_n + \beta_n^2 \right\}. & (30)
\end{aligned}$$

Notice that in (30) the function is quadratic $y = Ax^2 + Bx + C$ $(A > 0)$ in terms of β_n, so its minimum is obtained at

$$\widehat{\beta}_n := -\frac{B}{2A} = \frac{\frac{2}{n}\sum_{i=1}^n y_i}{2} := \bar{y}_n. \tag{31}$$

2.2 Part B

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(1)

Proof. In the regression model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + e_i$, we have

$$\widehat{\boldsymbol{y}} := \boldsymbol{X}\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1}\boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{H}\boldsymbol{y} \tag{32}$$

where $\boldsymbol{H}$ is the hat matrix, also by a decomposition we have $\boldsymbol{M} = \mathbb{I} - \boldsymbol{H}$ as the annihilator matrix, so a rearranging yields

$$\boldsymbol{y} = (\boldsymbol{H} + \boldsymbol{M})\boldsymbol{y} = \widehat{\boldsymbol{y}} + \underbrace{\boldsymbol{M}\boldsymbol{y}}_{\widehat{\boldsymbol{e}}} \tag{33}$$

which means we have the form $\boldsymbol{y} = \widehat{\boldsymbol{y}} + \widehat{\boldsymbol{e}}$. □

(2)

Proof. By our construction in **(1)**, we see that $\widehat{\boldsymbol{e}} \in Im(\boldsymbol{M})$, $\widehat{\boldsymbol{y}} \in Im(\boldsymbol{H})$. We next claim that $Im(\boldsymbol{M}), Im(\boldsymbol{H})$ are orthogonal to each other, hence $\widehat{\boldsymbol{y}}^\top \widehat{\boldsymbol{e}} = 0$ (inner product are zero). □

(3)

Proof. If intercept is included, wlog we let $\boldsymbol{x}_i = (1, x_{i2}, \cdots, x_{ip})^\top$. We claim that $\bar{\boldsymbol{y}} := (\bar{y}, \cdots, \bar{y})^\top \in Im(\boldsymbol{H})$, hence by linearity $\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}} \in Im(\boldsymbol{H})$ hence the inner product will yield zero: $\langle \widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}, \widehat{\boldsymbol{e}} \rangle = 0$. □

(4)

Proof. Note that we have

$$\begin{aligned} ||\boldsymbol{y} - \bar{\boldsymbol{y}}||_2^2 &= ||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}} + \boldsymbol{y} - \widehat{\boldsymbol{y}}||_2^2 &(34)\\ &= ||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}||_2^2 + ||\boldsymbol{y} - \widehat{\boldsymbol{y}}||_2^2 + 2\langle \widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}, \boldsymbol{y} - \widehat{\boldsymbol{y}} \rangle &(35) \end{aligned}$$

where by definition $\widehat{e} := \boldsymbol{y} - \widehat{\boldsymbol{y}} \in Im(\boldsymbol{M})$, we also have that $\boldsymbol{y} - \bar{\boldsymbol{y}} \in Im(\boldsymbol{H})$, and hence the inner product will yield zero: $\langle \widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}, \boldsymbol{y} - \bar{\boldsymbol{y}} \rangle = 0$, where we have

$$||\boldsymbol{y} - \bar{\boldsymbol{y}}||_2^2 := ||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}||_2^2 + ||\widehat{e}||_2^2. \tag{36}$$

□

Proof. Observe that

$$R^2 = \frac{\sum_{i\in[n]}(\widehat{y}_i - \bar{y})^2}{\sum_{i\in[n]}(y_i - \bar{y})^2} = \frac{||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}||_2^2}{||\boldsymbol{y} - \bar{\boldsymbol{y}}||_2^2} = \frac{||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}||_2^2}{||\widehat{\boldsymbol{y}} - \bar{\boldsymbol{y}}||_2^2 + ||\widehat{e}||_2^2} \in [0, 1] \tag{37}$$

since the norm is non-negative. □

3 Exercise 3

3.1 Part A

Notations: $\boldsymbol{x}_i = (x_{i1}, \cdots, x_{ip})^\top, i \in [n]$ *is the covariate vector,* $\boldsymbol{y} = (y_1, \cdots, y_n)^\top$ *is the response,* $\boldsymbol{\beta} = (\beta_1, \cdots, \beta_p)^\top$ *is the parameter vector,* $\boldsymbol{X}$ *is the design matrix.*

(1)

Proof. By definition, we have

$$\widehat{\boldsymbol{\beta}}^{(i)} := \arg\min_{\boldsymbol{u}\in\mathbb{R}^p} \frac{1}{n} \sum_{j\in[n]\setminus\{i\}} \{y_i - \boldsymbol{x}_i^\top \boldsymbol{u}\}^2. \tag{38}$$

We have seen in class that, the solution to (38) takes the form

$$\sum_{j\in[n]\setminus\{i\}} \boldsymbol{x}_j \boldsymbol{x}_j^\top \widehat{\boldsymbol{\beta}}^{(i)} = \sum_{j\in[n]\setminus\{i\}} \boldsymbol{x}_j y_j \tag{39}$$

by some modifications, we can rewrite the equation above as

$$\left(\boldsymbol{X}^\top \boldsymbol{X} - \boldsymbol{x}_i \boldsymbol{x}_i^\top\right) \cdot \widehat{\boldsymbol{\beta}}^{(i)} = \left(\boldsymbol{x}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i\right), \tag{40}$$

We will introduce Sherman-Morrison inversion formula first:

Sherman-Morrison Inversion Formula

Suppose $A \in M_n(\mathbb{R})$ be invertible and $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$, then $A + \boldsymbol{u}\boldsymbol{v}^\top$ is invertible if and only if $1 + \boldsymbol{u}^\top A^{-1} \boldsymbol{u} \neq 0$ and the inverse is given by

$$(A + \boldsymbol{u}\boldsymbol{v}^\top)^{-1} = A^{-1} - \frac{A^{-1}\boldsymbol{u}\boldsymbol{v}^\top A^{-1}}{1 + \boldsymbol{v}^\top A^{-1} \boldsymbol{u}} \tag{41}$$

Note that $(\boldsymbol{X}^\top \boldsymbol{X} - \boldsymbol{x}_i \boldsymbol{x}_i^\top)$ is clearly invertible since first $\boldsymbol{X}^\top \boldsymbol{X}$ is positive definite. To compute the inverse, we apply the formula directly:

$$\begin{aligned}
\widehat{\boldsymbol{\beta}}^{(i)} &:= \left(\boldsymbol{X}^\top \boldsymbol{X} - \boldsymbol{x}_i \boldsymbol{x}_i^\top\right)^{-1} \cdot \left(\boldsymbol{x}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i\right) && (42) \\
&= \left[\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} + \frac{(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}}{1 - \boldsymbol{x}_i^\top (\boldsymbol{X}\boldsymbol{X})^{-1} \boldsymbol{x}_i}\right] \cdot (\boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i) && (43) \\
&= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y} - \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{x}_i y_i + \frac{(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}}{1 - \boldsymbol{x}_i^\top (\boldsymbol{X}\boldsymbol{X})^{-1} \boldsymbol{x}_i} \cdot (\boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i) && (44) \\
&= \widehat{\boldsymbol{\beta}} + \frac{(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \cdot (\boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i) - (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i y_i (1 - h_{ii})}{1 - h_{ii}} && (45)
\end{aligned}$$

We denote the numerator of the fraction in (45) as $(*)$, then by some simplifications we have

$$(*) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}_i \cdot \left(\boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \cdot (\boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{x}_i y_i) - y_i(1-h_{ii})\right) \tag{46}$$

$$= (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}_i \cdot \left(\boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y} - \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}_i y_i - y_i + y_i \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}\right) \tag{47}$$

$$= (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}_i \cdot \left(\boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y} - y_i\right) \tag{48}$$

$$:= (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{x}_i \cdot (\widehat{y}_i - y_i) \tag{49}$$

Hence we may express $\widehat{\boldsymbol{\beta}}^{(i)}$ as

$$\widehat{\boldsymbol{\beta}}^{(i)} := \boldsymbol{\beta} - \frac{\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1}\boldsymbol{x}_i \widehat{e}_i}{1-h_{ii}}. \tag{50}$$

where $\widehat{e}_i = y_i - \widehat{y}_i$.

□

(2)

Using (50), we have

$$\widehat{e}_i^{(i)} := y_i - \boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}^{(i)} \tag{51}$$

$$= y_i - \boldsymbol{x}_i^\top \left[\boldsymbol{\beta} - \frac{\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1}\boldsymbol{x}_i \widehat{e}_i}{1-h_{ii}}\right] \tag{52}$$

$$= y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} + \frac{\boldsymbol{x}_i^\top \left(\boldsymbol{X}^\top \boldsymbol{X}\right)\boldsymbol{x}_i \widehat{e}_i}{1-h_{ii}} \tag{53}$$

$$= y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} + \frac{h_{ii}\widehat{e}_i}{1-h_{ii}} \tag{54}$$

$$\overset{\text{by (49)}}{=} y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} + \frac{h_{ii}}{1-h_{ii}} \cdot \left(y_i - \widehat{y}_i\right) \tag{55}$$

$$:= (y_i - \widehat{y}_i) + \frac{h_{ii}}{1-h_{ii}} \cdot (y_i - \widehat{y}_i) \tag{56}$$

$$= \frac{1}{1-h_{ii}} \cdot (y_i - \widehat{y}_i). \tag{57}$$

(3)

Using (50) and the definition of $\widehat{\mathcal{L}}_{loo}$, we have

$$\widehat{\mathcal{L}}_{loo}(\widehat{m}_n(\boldsymbol{x})) = \frac{1}{n}\sum_{i\in[n]} \left\{ y_i - \boldsymbol{x}_i^\top \left(\boldsymbol{\beta} - \frac{\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1}\boldsymbol{x}_i \widehat{e}_i}{1-h_{ii}}\right)\right\}^2 \tag{58}$$

$$= \frac{1}{n}\sum_{i\in[n]} \left\{ y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - \frac{\boldsymbol{x}_i^\top \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1}\boldsymbol{x}_i \widehat{e}_i}{1-h_{ii}}\right\}^2 \tag{59}$$

where we have $\boldsymbol{x}_i^\top \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{x}_i := h_{ii}$ so we now have

$$\widehat{\mathcal{L}}_{loo}(\widehat{m}_n(\boldsymbol{x})) = \frac{1}{n} \sum_{i\in[n]} \left\{ y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - \frac{h_{ii}}{1-h_{ii}} \widehat{e}_i \right\}^2. \tag{60}$$

Define $e_i = y_i - \boldsymbol{x}^\top \boldsymbol{\beta}$, then

$$\begin{aligned} \widehat{\mathcal{L}}_{loo}(\widehat{m}_n(\boldsymbol{x})) &= \frac{1}{n} \sum_{i\in[n]} \left\{ e_i - \frac{h_{ii}}{1-h_{ii}} \widehat{e}_i \right\}^2 && (61) \\ &= \frac{1}{n} \sum_{i\in[n]} \left\{ e_i^2 - \frac{2h_{ii}}{1-h_{ii}} e_i \widehat{e}_i + \left(\frac{h_{ii}}{1-h_{ii}} \cdot \widehat{e}_i \right)^2 \right\} && (62) \end{aligned}$$

where we have defined

$$e_i = y_i - \boldsymbol{x}^\top \boldsymbol{\beta}. \tag{63}$$

3.2 Part B

(1)

Proof. By direct computation, we have

$$\begin{aligned} \mathcal{L}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x})) &= \mathbb{E} \left\{ y^2 - 2y\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}} + (\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}})(\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}}) \right\} && (64) \\ &= \mathbb{E}[y^2] - 2\mathbb{E}[y\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}}] + \mathbb{E}[(\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}})^2]. && (65) \end{aligned}$$

For the first term in (65), we use $y = \boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon$ and hence

$$\begin{aligned} \mathbb{E}[y^2] &= \mathbb{E}\left[(\boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon)^2 \right] && (66) \\ &= \mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})^2] + 2\mathbb{E}[\boldsymbol{x}^\top \boldsymbol{\beta} \epsilon] + \mathbb{E}[\epsilon^2] && (67) \\ &= \mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})^2] + 0 + \mathbf{Var}(\epsilon) && (68) \\ &:= \Big(\mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})] \Big)^2 + \sigma^2 && (69) \end{aligned}$$

since $\mathbb{E}[\epsilon] = 0$ so $\mathbb{E}[g(\boldsymbol{x})\epsilon] = 0$, and we use the fact that

$$\mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})^2] = \underbrace{\mathbf{Var}(\boldsymbol{x}^\top \boldsymbol{\beta})}_{:=0} + \Big(\mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})] \Big)^2 \tag{70}$$

For the second term in (65), we have

$$\begin{aligned} -2\mathbb{E}[y\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}}] &= -2\mathbb{E}[y] \cdot \mathbb{E}[\boldsymbol{x}^\top \widehat{\boldsymbol{\beta}}] && (71) \\ &= -2\mathbb{E}[\boldsymbol{x}^\top \boldsymbol{\beta} + \epsilon] \cdot \mathbb{E}[\boldsymbol{x}^\top \boldsymbol{\beta}] && (72) \\ &= -2\Big(\mathbb{E}[(\boldsymbol{x}^\top \boldsymbol{\beta})] \Big)^2 && (73) \end{aligned}$$

using the fact that $\widehat{\boldsymbol{\beta}}$ is (conditionally) unbiased. Finally for the third term in (65), we have

$$\mathbb{E}[(\boldsymbol{x}^\top\widehat{\boldsymbol{\beta}})^2] = \mathbf{Var}(\boldsymbol{x}^\top\widehat{\boldsymbol{\beta}}) + \Big(\mathbb{E}[(\boldsymbol{x}^\top\widehat{\boldsymbol{\beta}})]\Big)^2 \tag{74}$$

$$= \boldsymbol{x}^\top \cdot \sigma^2\Big(\boldsymbol{X}^\top\boldsymbol{X}\Big)^{-1}\boldsymbol{x} + \Big(\mathbb{E}[\boldsymbol{x}^\top\beta]\Big)^2 \tag{75}$$

$$= \sigma^2 h_{xx} + \Big(\mathbb{E}[\boldsymbol{x}^\top\beta]\Big)^2 \tag{76}$$

where we use the fact that $\boldsymbol{x}$ is fixed. Combine with (99),(102),(105), we have the desired result:

$$\mathcal{L}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x})) = \sigma^2 + \sigma^2 h_{xx} = \sigma^2(1+h_{xx}). \tag{77}$$

□

(2)

Proof. To compute

$$\mathbb{E}\left\{\frac{1}{n}\sum_{i\in[n]}(y_i - \boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}})^2\right\} = \mathbb{E}\left\{\frac{1}{n}\sum_{i\in[n]}\Big(y_i^2 - 2y_i\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}} + (\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}})^2\Big)\right\}, \tag{78}$$

note that the general idea is similar to **(1)**, where the first and last term will yield

$$\mathbb{E}\left\{\frac{1}{n}\sum_{i\in[n]}\Big(y_i^2 + (\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}})^2\Big)\right\} = \frac{1}{n}\sum_{i=1}^{n}\Big\{\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta}+\epsilon)^2 + \mathbf{Var}(\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}) + \Big[\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})\Big]^2\Big\} \tag{79}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\Big\{2\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})^2 + \mathbf{Var}(\epsilon) + \boldsymbol{x}_i^\top\mathbf{Var}(\widehat{\boldsymbol{\beta}})\boldsymbol{x}_i\Big\} \tag{80}$$

$$= \left\{\frac{2}{n}\sum_{i=1}^{n}\Big(\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})\Big)^2\right\} + \sigma^2 + \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i^\top\sigma^2\Big(\boldsymbol{X}^\top\boldsymbol{X}\Big)^{-1}\boldsymbol{x}_i \tag{81}$$

$$= \sigma^2 + \frac{p}{n}\sigma^2 + \frac{2}{n}\sum_{i=1}^{n}\Big(\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})\Big)^2. \tag{82}$$

On the other hand, we have

$$\frac{2}{n}\sum_{i=1}^{n}\mathbb{E}[\boldsymbol{y}_i\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}] = \frac{2}{n}\sum_{i=1}^{n}\mathbb{E}\Big[(\boldsymbol{x}_i^\top\boldsymbol{\beta}+e_i)\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}\Big] \tag{83}$$

$$= \frac{2}{n}\sum_{i=1}^{n}\Big\{\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})^2 + \mathbb{E}[e_i\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}]\Big\} \tag{84}$$

$$= \frac{2}{n}\sum_{i=1}^{n}\Big(\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})\Big)^2. \tag{85}$$

Thus by combining (82),(85), one have

$$\widehat{\mathcal{L}}_{\text{naïve}} = \frac{1}{n}\sum_{i=1}^{n}\Big(y_i - \boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}\Big)^2 = \sigma^2 + \frac{p}{n}\cdot\sigma^2, \tag{86}$$

hence

$$\begin{aligned}
\text{Bias}\left[\mathcal{L}_{\text{naïve}}\right] &= \mathbb{E}\left[\widehat{\mathcal{L}}_{\text{naïve}}\right] - \mathcal{L}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x})) && (87)\\
&= \sigma^2 + \frac{p}{n}\cdot\sigma^2 - \sigma^2(1+h_{xx}) && (88)\\
&= \sigma^2\cdot\left(\frac{p}{n} - h_{xx}\right). && (89)
\end{aligned}$$

□

(3)

The idea is similar, but we first claim that $\widehat{\boldsymbol{\beta}}(E)$ is conditionally unbiased, since one have

$$\widehat{\boldsymbol{\beta}}(E) = \left(\sum_{i\in E}\boldsymbol{x}_i\boldsymbol{x}_i^\top\right)\cdot\left(\sum_{i\in E}\boldsymbol{x}_i y_i\right) := \left(\boldsymbol{X}_E^\top\boldsymbol{X}_E\right)^{-1}\cdot\boldsymbol{X}_E^\top\boldsymbol{y}_E \tag{90}$$

then it follows that

$$\begin{aligned}
\mathbb{E}\left[\widehat{\boldsymbol{\beta}}(E)\middle|\boldsymbol{X}\right] &= \mathbb{E}\left[\widehat{\boldsymbol{\beta}}(E)\middle|\boldsymbol{X}_E\right] && (91)\\
&= \left(\boldsymbol{X}_E^\top\boldsymbol{X}_E\right)^{-1}\cdot\boldsymbol{X}_E^\top\mathbb{E}\left[\boldsymbol{y}_E\middle|\boldsymbol{X}_E\right] && (92)\\
&= \left(\boldsymbol{X}_E^\top\boldsymbol{X}_E\right)^{-1}\cdot\boldsymbol{X}_E^\top\mathbb{E}\left[\boldsymbol{X}_E\boldsymbol{\beta}+\boldsymbol{\epsilon}\middle|\boldsymbol{X}_E\right] && (93)\\
&= \boldsymbol{\beta}. && (94)
\end{aligned}$$

Similarly we claim it also have the property that $\mathbf{Var}(\widehat{\boldsymbol{\beta}}(E)) := \sigma^2\left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}$. We now first compute the expectation of the testing estimator:

$$\begin{aligned}
\mathbb{E}[\widehat{\mathcal{L}}_{\text{test}}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x}))] &= \mathbb{E}\left[\frac{1}{n-n_T}\sum_{i\in E^c}\left(y_i - \boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)\right)^2\right] && (95)\\
&= \frac{1}{n-n_T}\sum_{i\in E^c}\left\{\mathbb{E}[y_i^2] - 2\mathbb{E}\left[y_i\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)\right] + \mathbb{E}\left[\left(\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)\right)^2\right]\right\} && (96)
\end{aligned}$$

We use $\boldsymbol{y}_i = \boldsymbol{x}_i^\top\boldsymbol{\beta} + \epsilon_i$ where $\boldsymbol{\beta}$ is the true parameter, so we will have

$$\begin{aligned}
\mathbb{E}[y_i^2] &= \mathbb{E}[(\boldsymbol{x}_i^\top\boldsymbol{\beta})^2] + 2\mathbb{E}[\boldsymbol{x}_i^\top\boldsymbol{\beta}\epsilon_i] + \mathbb{E}[\epsilon_i^2] && (97)\\
&= \mathbb{E}[(\boldsymbol{x}_i^\top\boldsymbol{\beta})^2] + \sigma^2 && (98)\\
&= \cancel{\mathbf{Var}(\boldsymbol{x}_i^\top\boldsymbol{\beta})} + \left(\mathbb{E}(\boldsymbol{x}_i^\top\boldsymbol{\beta})\right)^2 + \sigma^2 && (99)
\end{aligned}$$

$$\begin{aligned}
-2\mathbb{E}\left[y_i\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)\right] &= -2\mathbb{E}[y_i]\cdot\mathbb{E}[\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)] && (100)\\
&= -2\mathbb{E}[\boldsymbol{x}_i^\top\boldsymbol{\beta}+\epsilon_i]\cdot\mathbb{E}[\boldsymbol{x}_i^\top\widehat{\boldsymbol{\beta}}(E)] && (101)\\
&= -2\left(\mathbb{E}[\boldsymbol{x}_i^\top\boldsymbol{\beta}]\right)^2 && (102)
\end{aligned}$$

$$\mathbb{E}\Big[\Big(\boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}(E)\Big)^2\Big] = \mathbf{Var}(\boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}(E)) + \Big(\mathbb{E}[\boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}(E)]\Big)^2 \tag{103}$$

$$= \boldsymbol{x}_i^\top \mathbf{Var}(\widehat{\boldsymbol{\beta}}(E))\boldsymbol{x}_i + \Big(\mathbb{E}[\boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}(E)]\Big)^2 \tag{104}$$

$$= \boldsymbol{x}_i^\top \mathbf{Var}(\widehat{\boldsymbol{\beta}}(E))\boldsymbol{x}_i + \Big(\mathbb{E}[\boldsymbol{x}_i^\top \boldsymbol{\beta}]\Big)^2 \tag{105}$$

The by (99),(102),(105), we have

$$\mathbb{E}[\widehat{\mathcal{L}}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x}))] = \frac{1}{n-n_T}\sum_{i\in E^c}\Big\{\mathbb{E}[(\boldsymbol{x}_i^\top \boldsymbol{\beta})^2] + \sigma^2 - 2\Big(\mathbb{E}[\boldsymbol{x}^\top \boldsymbol{\beta}]\Big)^2 + \boldsymbol{x}_i^\top \mathbf{Var}(\widehat{\boldsymbol{\beta}}(E))\boldsymbol{x}_i + \Big(\mathbb{E}[\boldsymbol{x}_i^\top \widehat{\boldsymbol{\beta}}(E)]\Big)^2\Big\} \tag{106}$$

$$= \frac{1}{n-n_T}\sum_{i\in E^c}\Big\{\sigma^2 + \boldsymbol{x}_i^\top \mathbf{Var}(\widehat{\boldsymbol{\beta}}(E))\boldsymbol{x}_i\Big\} \tag{107}$$

Then the bias is given by

$$\text{Bias}\Big[\widehat{\mathcal{L}}_{\text{test}}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x})\Big] = \mathbb{E}\Big[\widehat{\mathcal{L}}_{\text{test}}(\widehat{\boldsymbol{m}}_n(\boldsymbol{x})\Big] - \mathcal{L}(\boldsymbol{m}_n(\boldsymbol{x})) \tag{108}$$

$$= \frac{1}{n-n_T}\sum_{i\in E^c}\Big\{\sigma^2 + \boldsymbol{x}_i^\top \mathbf{Var}(\widehat{\boldsymbol{\beta}}(E))\boldsymbol{x}_i\Big\} - \sigma^2(1+h_{xx}) \tag{109}$$

$$= \frac{1}{n-n_T}\sum_{i\in E^c}\Big\{\sigma^2 + h_{ii}\Big\} - \sigma^2(1+h_{xx}) \tag{110}$$

$$= \sigma^2\left(\frac{1}{n-n_T}\sum_{i\in E^c} h_{ii} - h_{xx}\right). \tag{111}$$

(4)

We use the similar idea from **(3)**, and we use the property that the estimator is conditionally unbiased with covariance matrix $\sigma^2\Big(\boldsymbol{X}^\top \boldsymbol{X}\Big)^{-1}$. The similar computation leads to

$$\text{Bias}\Big[\widehat{\mathcal{L}}_{\text{loo}}(\widehat{\boldsymbol{m}}(\boldsymbol{x}))\Big] = \frac{1}{n}\sum_{i\in[n]}\Big(\sigma^2 + \boldsymbol{x}_i \mathbf{Var}(\widehat{\boldsymbol{\beta}}^{(i)})\boldsymbol{x}_i\Big) - \sigma^2(1+h_{xx}) \tag{112}$$

$$= \frac{1}{n}\sum_{i\in[n]}\Big(\sigma^2 + \sigma^2 h_{ii}\Big) - \sigma^2(1+h_{xx}) \tag{113}$$

$$= \sigma^2 + \sigma^2\cdot\frac{p}{n} - \sigma^2 - \sigma^2 h_{xx} \tag{114}$$

$$= \sigma^2\left(\frac{p}{n} - h_{xx}\right). \tag{115}$$

(5)

I don't really have much time for this one...

4 Coding Part

4.1 Part B

In this task, we examine the `mtcars` dataset in **R**. The dataset contains information about fuel consumption and design aspects of 32 car models from 1974. The response variable of interest is `mpg` (miles per gallon), representing fuel efficiency while other parameters are considered as covaraites. The dataset includes the following variables:

Variable	Description
`mpg`	Miles per gallon (response)
`wt`	Weight of the car
`hp`	Horsepower
`am`	Transmission (0 = automatic, 1 = manual)
`cyl`	Number of cylinders
`disp`	Engine displacement (cu.in.)

The following command can be used to access the first few samples:

```
> head(mtcars)
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

We aim to answer the following question: How do a car's weight, horsepower, and transmission type influence its fuel efficiency? We fit the following model:

$$\texttt{mpg} = \beta_0 + \beta_1\texttt{wt} + \beta_2\texttt{hp} + \beta_3\texttt{am} + \varepsilon \tag{116}$$

```
model <- lm(mpg ~ wt + hp + am, data = mtcars)
summary(model)
```

Listing 1: Fitting the linear regression model

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 34.00     2.64       12.9    <0.001 ***
wt          -2.88     0.74       -3.9    0.001 ***
hp          -0.037    0.009      -4.1    <0.001 ***
am           2.08     1.38        1.5    0.14
R-squared = 0.83
```

- **Weight** (`wt`) and **horsepower** (`hp`) have significant negative effects on `mpg`.

- **Transmission** (`am`) shows higher fuel efficiency for manual cars, though not statistically significant at 5%.
- The model explains roughly 83% of the variability in `mpg`.

```
plot(mtcars$mpg, fitted(model),
     xlab="Observed MPG", ylab="Predicted MPG",
     main="Observed vs Predicted Fuel Efficiency")
abline(0,1,col="red")
```

Listing 2: Observed vs predicted MPG

And the plot is given by the figure below:

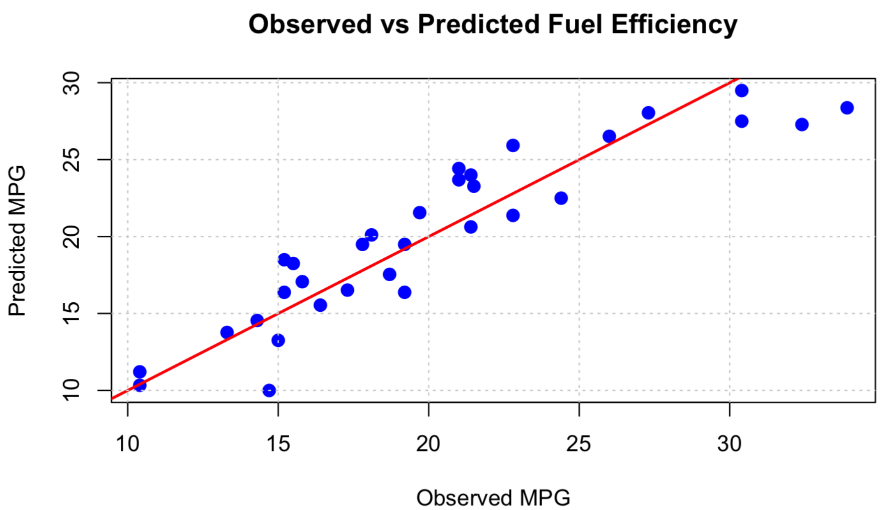

Figure 1: The plot we have